FIN Laboratory Transfer Package — Release 10.21

P240 Optimal Tomography + P241 Blind-Custody Validator + P242 One-Shot Pipeline

A scientific transfer package and executable specification
for theoretical and experimental physicists

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Transfer target. One finite, calibrated, twelve-state laboratory test of the strict FIN generator, with an independently signed event bundle, a sealed 2τ holdout and exactly one registered analysis.

Central mathematical object.

$$W = K_s, \qquad s = 1.660307278766099, \qquad A = sI - W \geq 0, \qquad U_t = e^{-itA}, \quad P_t = e^{-tA}.$$

Critical boundary. This package supplies a design, validator and analysis pipeline. It supplies no external events, physical clock, apparatus, strict selector, legacy–strict role transfer, L_{total}, Standard Model/gravity derivation or Theory-of-Everything closure.

Repository. `https://github.com/hyconiek/Fractal-Nadsoliton-Theory`
Related project DOI supplied by the author. `https://doi.org/10.5281/zenodo.21435332`

Transfer statement

This document is intended to be handed, together with the hashed executable files, to a laboratory team. It summarizes the finite spectral theory, the wave–diffusion duality, the missing operational object, the three candidate apparatus classes, the registered acquisition design, the independent custody contract and the one-shot analysis rule.

Programs 240–242 in this document are laboratory programme numbers from the Release-10.20 roadmap. They are not the older internal FAR packets that happen to contain the same integers.

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0.1 Confidence convention

Analytic proof, machine checking, synthetic planning, feasibility hypotheses, open obligations and scoped refutations are kept separate throughout this transfer package.

Chapter 1

Abstract

This document transfers the experimentally actionable part of the FIN research programme to an independent team of theoretical and experimental physicists. It begins from the finite strict twelve-node kernel, not from cosmological or Theory-of-Everything claims. With

$$W = K_s, \qquad s = 1.660307278766099, \qquad A = sI - W,$$

the exact Dirichlet identity

$$\langle f, Af \rangle = \frac{1}{2} \sum_{x,y} W_{xy} |f_x - f_y|^2 \geq 0$$

places the same self-adjoint generator under the coherent group

$$U_t = e^{-itA}$$

and the reversible heat/Markov semigroup

$$P_t = e^{-tA}.$$

This duality is mathematically proved on the frozen C_{12} target. It does not select which temporal law is physical. State space, preparation, clock, instrument, environment, apparatus, record and dimensional calibration remain operational inputs. The document therefore does not ask a laboratory to confirm a metaphysical interpretation. It asks whether an explicitly realized, calibrated and blinded twelve-state process follows the registered FIN transition law and whether it survives a one-time held-out semigroup test.

Program 240 is executed here. The fastest-mode inverse-log noise factor is e^x/x, with the unique optimum

$$x = \tau \lambda_{\max} = 1.$$

Cyclic symmetrization makes equal allocation to all twelve basis preparations minimax within the declared convex design class. A rectangular matrix-Bernstein theorem supplies a nonasymptotic spectral concentration bound. Synthetic planning gives a 99% and 100% pass rate for the frozen projective spectral gate at respectively 25 000 and 50 000 shots per preparation, while the fully distribution-free sufficient count for the same 0.02 projective bound is approximately 22.6 million per preparation. The large difference is reported, not hidden.

Program 241 is an executable blind-custody validator. It checks eleven measurement, data, calibration, control, chronology, hash and role-separation fields and requires a detached signature from a trusted registrar. It accepts either a twelve-state process record or an event-level double-slit record. It cannot generate the missing independent empirical record.

Program 242 is a fail-closed analysis pipeline. It runs exactly once for an admitted twelve-state process bundle, predicts

$$P_{2\tau}^{\mathrm{pred}} = \widehat{P}_{\tau}^{2},$$

compares the prediction with the sealed $\widehat{P}_{2\tau}$ holdout, evaluates the strict projective fingerprint and preregistered negative controls, and reports failure without model repair. No external bundle is supplied in this release; consequently Program 242 remains locked and no physical validation is claimed.

Chapter 2

Executive decision for a laboratory team

2.1 What is ready

- The finite strict target matrix and spectrum are frozen.
- The dual unitary/heat mathematics is proved on that target.
- The heat-kernel instrument-to-spectrum theorem is explicit.
- P240 supplies the registered time, allocation, shot plan and concentration bounds.
- P241 supplies a production validator, transfer template and signed-custody gate.
- P242 supplies a one-run semigroup/fingerprint pipeline.
- Eighteen regression tests pass.

2.2 What is not ready

- No external event bundle exists in this package.
- No code can manufacture independent custody, an independent apparatus or a real physical preparation.
- No laboratory is asserted to realize the twelve basis states, vertex POVM, or a homogeneous FIN semigroup.
- No absolute physical time, energy, action or length unit follows from the dimensionless operator.
- The strict selector obstruction QW-2191 remains open.
- The legacy-to-strict completion bridge and legacy physical-role transfer remain open.
- No L_{total}, Standard Model, general-relativistic or Theory-of-Everything closure is claimed.

2.3 Platform recommendation

Platform	**Direct relevance to P227/ P240–P242**	**Recommended role**
P-A: twelve-mode photonic coherent walk plus separately verified stochastic/open-system realization	high	primary FIN process experiment
P-B: qubit Mach–Zehnder/ Ramsey single–plus–echo dephasing experiment	medium as an operational-method test; low as a twelve-node FIN test	auxiliary memory, dephasing and instrument validation

Platform	**Direct relevance to P227/ P240–P242**	**Recommended role**
P-C: event-level single-photon double slit with four shutter configurations	high for event acquisition and custody; low for direct 12×12 spectral reconstruction	procedural pilot and independent interference record

The recommended scientific sequence is:

1. use P-C, if convenient, to prove that the collaboration can produce the required event-level, signed, blinded record;
2. execute P-A as the direct twelve-state process test;
3. use P-B only for auxiliary tests of dephasing, echo, memory and detector modelling;
4. do not treat a programmable emulator as evidence that nature selected FIN; it validates the implementation and falsification pipeline.

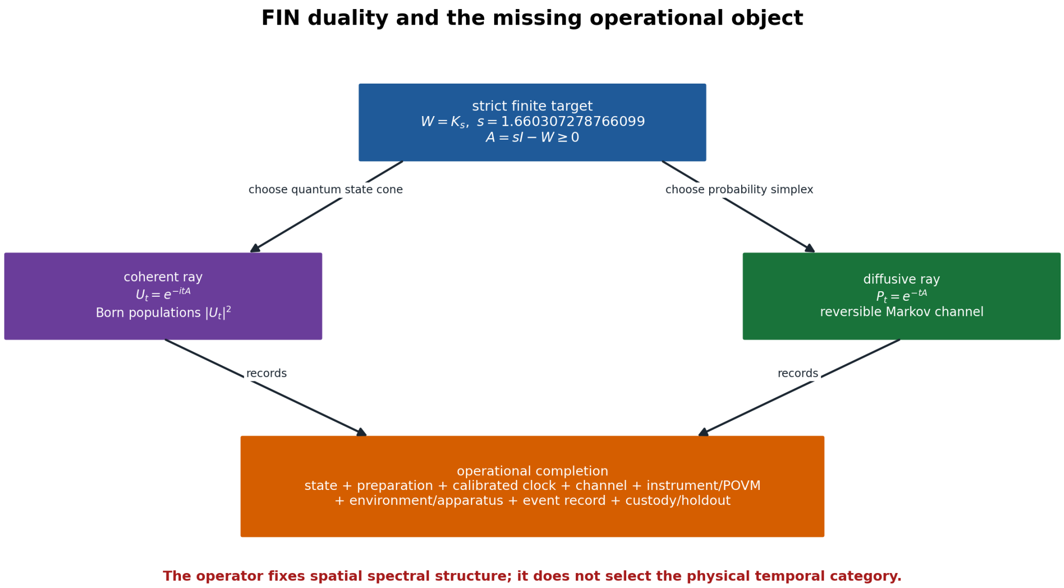

Figure 2.1: The same positive generator supports coherent and diffusive functional calculi, while empirical probabilities require an operational completion.

Chapter 3

Claim convention

Label	Meaning
Proven	analytic finite theorem in the declared scope
Machine checked	executable test or exact hash/schema check completed
Synthetic planning evidence	numerical experiment generated from a declared model; not physical evidence
Feasibility hypothesis	apparatus proposal requiring expert engineering review
Open	missing theorem, apparatus, calibration or external record
Refuted in scope	an explicit counterexample or no-go applies under stated assumptions

No statement is promoted across these categories.

Chapter 4

Scientific question

The laboratory question is intentionally narrower than the overall FIN research programme:

> *Can an independently operated, calibrated twelve-state platform realize the frozen transition family associated with the strict FIN generator, and can its sealed 2τ record be predicted from its calibration-time τ record without post-hoc repair?*

There are two associated but distinct questions:

1. does a coherent implementation realize the transition amplitudes of $U_t = e^{-itA}$;
2. does a stochastic or open-system implementation realize the populations of $P_t = e^{-tA}$?

The common generator makes these two calculations mathematically related. It does not make the two experimental procedures physically identical.

Chapter 5

Frozen finite target

5.1 Strict working kernel

Let $V = \mathbb{Z}/12\mathbb{Z}$ and

$$d(x, y) = \min\{|x - y|, 12 - |x - y|\}.$$

The frozen strict working profile is

$$K_{\text{strict}}(d) = \frac{\cos(0.18575\, d + 0.16250)}{1 + d^{1.8}}.$$

Define

$$W_{xx} = 0, \qquad W_{xy} = K_{\text{strict}}(d(x, y)) \quad (x \neq y).$$

The six positive distance weights are

d	$K_{\text{strict}}(d)$
1	0.46998567264502017
2	0.19204355169010282
3	0.09142861427792497
4	0.04702916874565040
5	0.02413122336363006
6	0.011070817321442113

Every row has sum

$$s = 2 \sum_{d=1}^{5} K_{\text{strict}}(d) + K_{\text{strict}}(6) = 1.6603072787660986,$$

which agrees with the registered value 1.660307278766099 to 4.44×10^{-16}.

5.2 Positive Dirichlet generator

Set

$$A = sI - W.$$

Because W is symmetric, nonnegative off diagonal and has constant row sum,

$$\begin{aligned}\langle f, Af\rangle &= s\sum_x |f_x|^2 - \sum_{x,y} \overline{f}_x W_{xy} f_y \\ &= \frac{1}{2}\sum_{x,y} W_{xy}|f_x - f_y|^2 \geq 0.\end{aligned}$$

Strict positivity of all off-diagonal weights makes

$$\ker A = \operatorname{span}\{\mathbf{1}\}.$$

The nonnegative generator eigenvalues, organized by Fourier mode, are:

Modes	$\lambda(A)$
0	0
±1	0.754121154207080
±2	1.577049514427610
±3	1.961406861976446
±4	2.199568849333210
±5	2.298606272079097
6	2.342182041146301

Thus

$$\lambda_{\max} = 2.342182041146298, \qquad \tau_* = \lambda_{\max}^{-1} = 0.426952295949885$$

in dimensionless FIN time.

5.3 Kernel policy

The laboratory target above is the later strict working kernel. It must not be silently identified with the canonical legacy intermediate kernel

$$K_{\text{legacy}}(d) = \alpha_{\text{geo}}\frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d}, \qquad \alpha_{\text{geo}} = 4\ln 2.$$

The current repository policy is:

- K_strict_gate is the primary strict finite target;
- K_legacy_ont is an intermediate bridge kernel;

- no raw identity, continuum completion theorem or physical-role transfer is available;
- a fixed-support polynomial interpolation on C_{12} is not a universal completion law.

The laboratory must therefore label the implemented matrix explicitly and must not infer legacy electroweak, electromagnetic or gravity-role formulas from a strict-kernel fit.

Chapter 6

One operator, multiple mathematical dynamics

6.1 Coherent group

Self-adjointness gives

$$U_t = e^{-itA}, \qquad U_t^\dagger U_t = I, \qquad U_{t+u} = U_t U_u.$$

If the initial amplitude is ψ_0, then

$$\psi(t) = U_t \psi_0.$$

A vertex detector with effects $E_x = |x\rangle\langle x|$ yields Born populations

$$M_t(x|j) = |\langle x|U_t|j\rangle|^2.$$

The matrices M_t are doubly stochastic at each fixed time, but generally

$$M_{t+u} \neq M_t M_u.$$

The missing cross terms are the finite interference witness.

6.2 Diffusive semigroup

Because $Q = -A = W - sI$ has nonnegative off-diagonal entries and zero column and row sums,

$$P_t = e^{tQ} = e^{-tA}$$

is positive, symmetric, doubly stochastic and satisfies

$$P_{t+u} = P_t P_u.$$

Its stationary distribution is uniform. The spectral gap controls mixing:

$$\|P_t - \Pi\|_{2\to 2} = e^{-0.754121154207080\,t}, \qquad \Pi = \frac{1}{12}\mathbf{1}\mathbf{1}^{\mathsf{T}}.$$

6.3 Short-time discriminator

For a localized preparation j and $i \neq j$,

$$P_t(i|j) = W_{ij}t + O(t^2),$$

whereas

$$M_t(i|j) = W_{ij}^2 t^2 + O(t^4).$$

Classical escape is linear and coherent escape is quadratic. A freely fitted clock can make one snapshot look similar, so the experiment must use a calibrated clock and multiple times.

6.4 Wave, Green and variational shadows

The same spectral measure also defines, when typed correctly:

$$\cos(t\sqrt{A}), \qquad \frac{\sin(t\sqrt{A})}{\sqrt{A}}, \qquad (A + zI)^{-1}, \qquad e^{-tA}.$$

The quadratic functional

$$S[\phi] = \frac{1}{2}\phi^{\mathsf{T}} A\phi - J^{\mathsf{T}}\phi$$

has stationary equation

$$A\phi = J.$$

This reconstructs a Green equation from the operator. It does not supply a physical action unit, a source J, a Lorentzian continuum, gauge fields or a full field Lagrangian.

6.5 The observer issue

The two dynamics are not an observer-induced contradiction. They are two inequivalent temporal semantics of one spatial generator. The mathematical observer is an operationally specified apparatus:

$$\mathcal{P} = (A, \mathcal{S}, \rho_0, \gamma, \{\Phi_\tau\}, \{\mathcal{I}_{a|x}\}, F_{\text{app}}, \mathcal{R}).$$

The entries denote generator, state/probability space, preparation, clock calibration, channel, instruments, apparatus frame and persistent record. Removing any entry creates a concrete nonidentifiability:

Removed object	Failure
state cone / probability rule	unitary and stochastic models remain possible
preparation	the uniform state hides both population dynamics
calibrated clock	rate and time can be inversely rescaled
instrument	intermediate observation and coherent propagation are conflated
environment	different memory/decoherence laws share terminal marginals
apparatus frame	odd/orientation data can be erased
record	no empirical likelihood or holdout exists

This is why a fifth duty—an independent record and custody split—cannot be generated by the simulation code.

Chapter 7

Dimensional and selector boundary

The finite matrix A is dimensionless. Physical evolution would require, for example,

$$U_{\tau_{\text{phys}}} = \exp\left[-i\frac{E_0\tau_{\text{phys}}}{\hbar}A\right],$$

or

$$P_{\tau_{\text{phys}}} = \exp[-\kappa\tau_{\text{phys}}A].$$

Only $E_0\tau_{\text{phys}}/\hbar$ or $\kappa\tau_{\text{phys}}$ is visible to the dimensionless operator. The minimal conversion package remains

$$\text{CA} = (\ell_*, \tau_*, \hbar_*)$$

or another independent rank-three basis of length, time and action/energy calibrations. It is imported through calibrated standards, not derived from a new dimensionless scalar.

Reflection exchanges Fourier modes $+m$ and $-m$. A radial operator cannot select an absolute orientation. A laboratory can supply a chiral preparation, directed apparatus or signed reference. That is an operational sector resource, not a strict-core discharge of QW-2191.

Shannon information computed from detector probabilities is dimensionless. Thermodynamic entropy, work and heat additionally require k_B, a physical ensemble, temperature, Hamiltonian/bath and work/heat instruments. The experiment below does not silently make that conversion.

Chapter 8

The exact empirical target

8.1 Primary heat-process prediction

The calibration record estimates

$$\widehat{P}_{\tau}.$$

Without using the sealed 2τ data, Program 242 freezes

$$\widehat{P}_{2\tau}^{\mathrm{pred}} = \widehat{P}_{\tau}\widehat{P}_{\tau}.$$

The held-out statistic is

$$T_{\mathrm{CK}} = \max_{j} \mathrm{TV}\left(\widehat{P}_{2\tau}(\cdot|j), \widehat{P}_{\tau}^{2}(\cdot|j)\right).$$

This is a Chapman–Kolmogorov/time-homogeneity test. A failure rejects the registered semigroup realization at the declared level. A non-rejection does not prove that nature selected FIN.

8.2 Secondary projective spectral fingerprint

For exact heat data,

$$A = -\frac{1}{\tau}\log P_{\tau}.$$

If the physical clock scale is unknown but stable, transition eigenvalues

$$\mu_k = e^{-\tau\lambda_k}$$

still determine

$$\frac{\lambda_k}{\lambda_{\max}} = \frac{\log \mu_k}{\log \mu_{\min}}.$$

The frozen secondary distance is the maximum absolute deviation between the reconstructed positive eigenvalue ratios and the strict target. The raw gate is 0.02.

8.3 Coherent duality diagnostics

If a coherent arm is available, it should additionally measure:

1. localized escape at $\tau/4, \tau/2, \tau$;
2. $M_{2\tau} - M_\tau^2$ with and without an intermediate vertex measurement;
3. phase-sensitive interference, not populations alone;
4. apparatus-reflection and preparation-label controls.

The heat arm must not be inferred merely from loss of coherent visibility.

Chapter 9

SECTION C — PROPOSED EXPERIMENT AND APPARATUS

Chapter 10

Five mandatory measurement duties

The five duties from Program 227 and the Program-228 intake gate are:

No.	Duty	Current mathematical status	Can code supply it?
1	vertex-basis or informationally complete preparations	explicit instrument model and design theorem	code specifies; laboratory must realize
2	calibrated duration τ	exact dimensionless optimum and calibration obligation	code specifies; external clock supplies units
3	vertex-resolving measurement / vertex POVM	exact instrument-to-spectrum theorem	code specifies; apparatus must realize and calibrate
4	finite transition counts	nonasymptotic and simulated shot analysis	code analyzes; detector supplies events
5	independent record, holdout and provider–registrar–analyst separation	schema, validator and one-run gate only	**no**; this field is empirically empty until humans supply it

Duties 1–4 have a theorem or a declared operational model. Duty 5 is not a software artifact and must not be filled with synthetic data, a screenshot, a rendered plot or a record produced by the same analysis team.

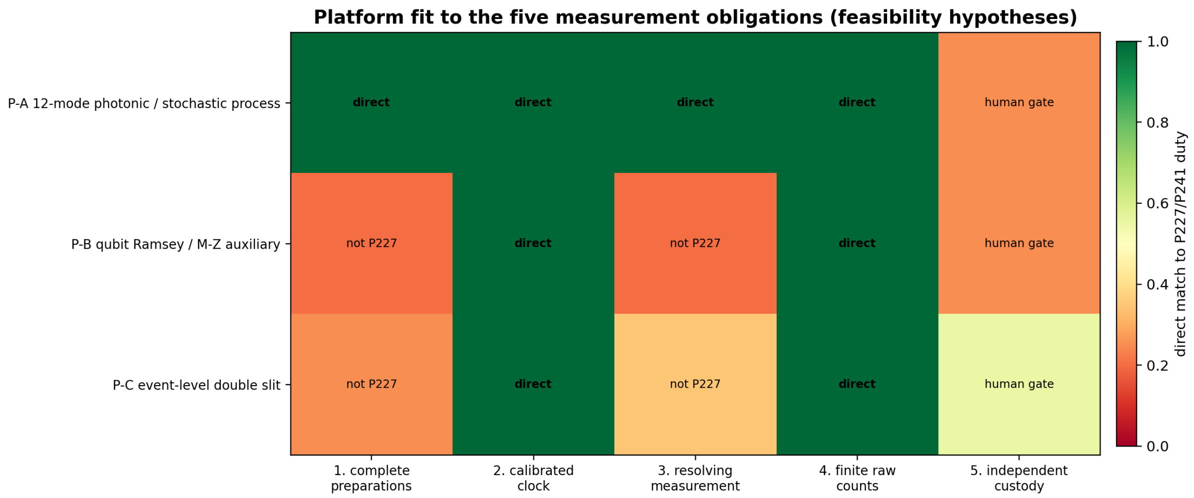

Figure 10.1: Fit of the three feasibility hypotheses to the five measurement duties. Independent custody remains a human and institutional gate.

Chapter 11

Platform P-A — twelve-mode photonic continuous-time walk

11.1 Feasibility status

Feasibility hypothesis, not a FIN theorem. Integrated waveguide arrays and programmable multiport interferometers can realize large coherent mode networks. Controlled noisy quantum walks and event-resolved photonic measurements also exist. These precedents make a twelve-mode implementation plausible. They do not prove that a given chip realizes the FIN matrix.

11.2 Apparatus

- heralded single-photon source based on SPDC, or a strongly attenuated laser with documented photon statistics and a separate classical-intensity calibration;
- source trigger and time-tagger reference;
- twelve independently selectable input modes;
- either femtosecond-laser-written waveguides or a reconfigurable Mach–Zehnder mesh implementing the coherent target;
- calibrated phase shifters and coupling characterization;
- an engineered dissipative/stochastic realization with transition rates W_{xy}, for example explicit Lindblad jumps

  $$L_{xy} = \sqrt{W_{xy}}\,|x\rangle\langle y|,$$

  or a separately programmed stochastic router;
- twelve SPAD or SNSPD output channels;
- TCSPC/time-tagger electronics;
- power, temperature, vibration and phase-stability monitors;
- independent calibration paths for detector efficiency, dark counts, crosstalk and timing jitter;
- shuttered/dark and nearest-neighbour controls.

11.3 Critical correction to the proposed setup

Ordinary propagation loss, pure dephasing or uncontrolled disorder does not by itself prove

$$\dot{p} = -Ap.$$

A passive lossy optical amplitude generally follows a non-Hermitian amplitude equation, and detected intensities are squared amplitudes. They are not automatically the columns of $e^{-\tau A}$. The heat arm must therefore be derived from explicit jump/routing dynamics or independently reconstructed by process tomography.

11.4 Mapping to the five duties

Duty	P-A implementation
1	inject each of the twelve input ports in randomized order
2	calibrate propagation/coupling time or programmed transition duration against an external clock
3	twelve resolved output channels; characterize confusion/crosstalk matrix
4	retain every timestamped detector event and run identifier
5	provider exports signed raw bundle to an independent registrar; analyst receives sealed split only through protocol

11.5 Program-228 fields potentially closed

P-A can close event order, preparation labels, evolution-time labels, detector calibration, environmental monitoring, negative controls and a held-out target. Provider identity, license, independent registrar signature and genuine role separation are organizational duties.

11.6 Physical boundary

This report does not prove that any laboratory implements twelve ideal basis states, a vertex POVM, the exact W, or a homogeneous FIN semigroup. An experimental physicist must decide whether the hardware realizes the stated channel within measured uncertainty.

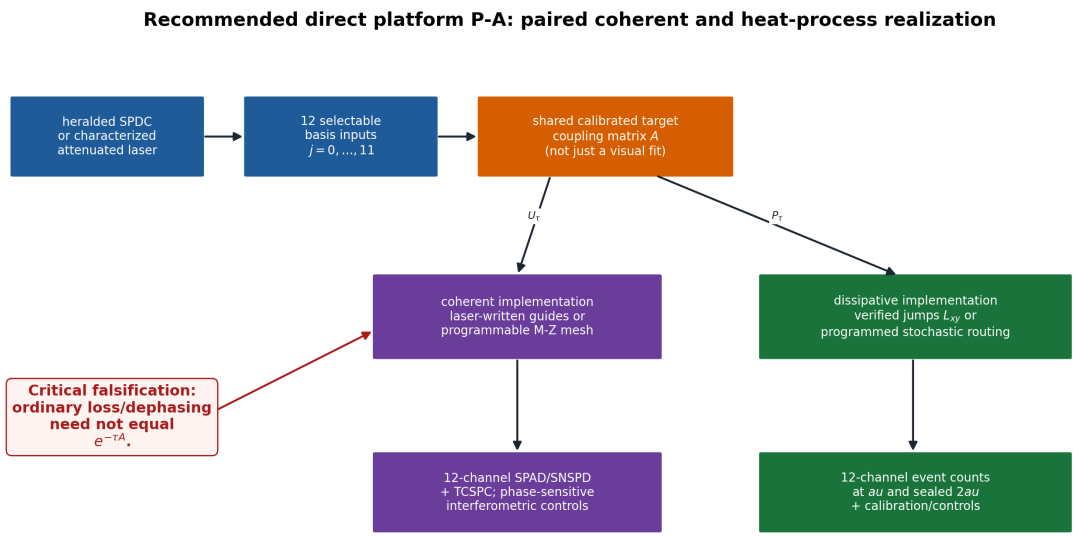

Figure 11.1: Recommended direct P-A architecture. Coherent propagation and heat flow require separately verified physical realizations of the common target generator.

Chapter 12

Platform P-B — qubit Mach–Zehnder/Ramsey dephasing

12.1 Feasibility status

Feasibility hypothesis and auxiliary protocol. A single qubit or two-path interferometer is suitable for the single/plus/echo identifiability problem and for testing detector blur, dephasing and temporal memory. It is not informationally complete for the twelve-node FIN generator.

12.2 Apparatus

- Mach–Zehnder interferometer, Ramsey-controlled qubit or equivalent two-level platform;
- coherent source or qubit initialization;
- AWG and microwave/optical pulse generation;
- EOM, piezo phase control or qubit Z control;
- single, plus and echo preparation/intervention sequences;
- calibrated readout and detector-confusion matrix;
- synchronized clock and event-level record;
- optional environment/noise monitor.

12.3 Role of the quoted $\tau_* = 0.002\,\mathrm{s}$

The value $0.002\,\mathrm{s}$ appeared in a conditional synthetic operational example. It is not a FIN-derived universal time and must not be imposed on a laboratory. The apparatus must select times from its independently calibrated coherence and control range.

12.4 What P-B can test

- whether single, plus and echo records identify declared dephasing and memory parameters;
- whether an intermediate intervention changes a multitime record;
- whether the apparatus blur model is stable on held-out runs;
- whether environment models with identical one-time channels differ at two times.

12.5 What P-B cannot establish

It cannot reconstruct twelve transition columns, the C_{12} projective fingerprint, or the P242 semigroup holdout without a new, explicit encoding of twelve preparations and twelve outcomes.

Chapter 13

Platform P-C — event-level double slit

13.1 Feasibility status

Feasibility hypothesis for an independent event record. Time-resolved single-photon double-slit measurements with SPAD arrays have been demonstrated in the literature. P-C is therefore a strong candidate for testing the custody, raw-event and interference-analysis procedures. It is not by itself a realization of the finite FIN C_{12} operator.

13.2 Apparatus

- heralded SPDC single-photon source or characterized attenuated source;
- spatial filtering and collimation;
- two-slit mask with independently actuated left/right shutters;
- four randomized configurations: both_open, left_only, right_only, both_closed;
- fixed propagation geometry and alignment/stability monitors;
- position-resolving EMCCD/ICCD or, preferably for direct timestamps, a SPAD array with TCSPC;
- source trigger/herald detector where applicable;
- independent dark-count, efficiency and point-spread calibration;
- raw event storage.

Each event row must contain at least:

event_id, timestamp_utc, run_id, subset, configuration, x_detector, y_detector, intervention

The raw record must be retained. A final image of interference fringes is insufficient.

13.3 Four-configuration analysis

Let $N_{BO}(x)$, $N_L(x)$, $N_R(x)$ and $N_{BC}(x)$ be efficiency- and exposure-corrected counts. An interference contrast can be formed from

$$C(x) = N_{BO}(x) - N_L(x) - N_R(x) + N_{BC}(x).$$

The contrast tests coherent cross terms relative to the registered controls. It does not identify A unless a twelve-mode mapping and propagation model were independently frozen.

13.4 Mapping to the five duties

Duty	P-C implementation
1	four slit configurations, not the twelve P227 basis preparations
2	timestamps and exposure/trigger calibration
3	spatially resolved detector, not automatically the twelve-outcome vertex POVM
4	event-by-event coordinate and time record
5	a real independent bundle can satisfy P241 when signed and custody-separated

13.5 Physical boundary

P-C can validate interference, background subtraction, raw-event storage and blind custody. A vanilla P-C bundle must make P242 return NOT_SEMIGROUP_READY. This is a safety feature.

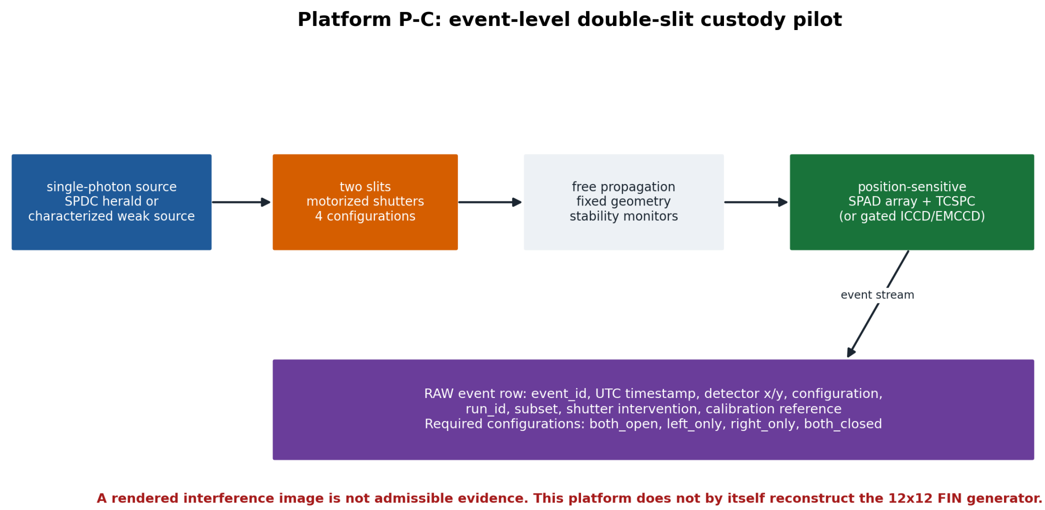

Figure 13.1: P-C event-level double-slit apparatus. Raw detections, four shutter configurations and calibrations are retained; a rendered image is not an admissible substitute.

Chapter 14

Blind-custody protocol

The roles are disjoint:

$$\text{provider} \neq \text{registrar} \neq \text{analyst}.$$

14.1 Provider

1. acquires raw events;
2. records all calibration, control and environment files;
3. does not see the frozen analysis result;
4. transfers the unmodified bundle to the registrar.

14.2 Registrar

1. verifies event and metadata completeness;
2. computes SHA-256 for every file;
3. freezes the bundle digest;
4. verifies that preregistration predates acquisition;
5. seals the holdout;
6. signs bundle_manifest.json with a detached GPG signature;
7. releases only the authorized split at each stage.

14.3 Analyst

1. freezes P240/P242 code and the analysis-lock hash before unblinding;
2. does not possess the registrar signing key;
3. receives the calibration subset first;
4. performs no threshold or model repair after seeing the holdout;
5. runs P242 once.

14.4 One-run rule

P242 creates an atomic ledger entry keyed by the bundle digest before reading the unblinded holdout. If an entry already exists, the pipeline refuses a second run. A failed run remains part of the scientific record.

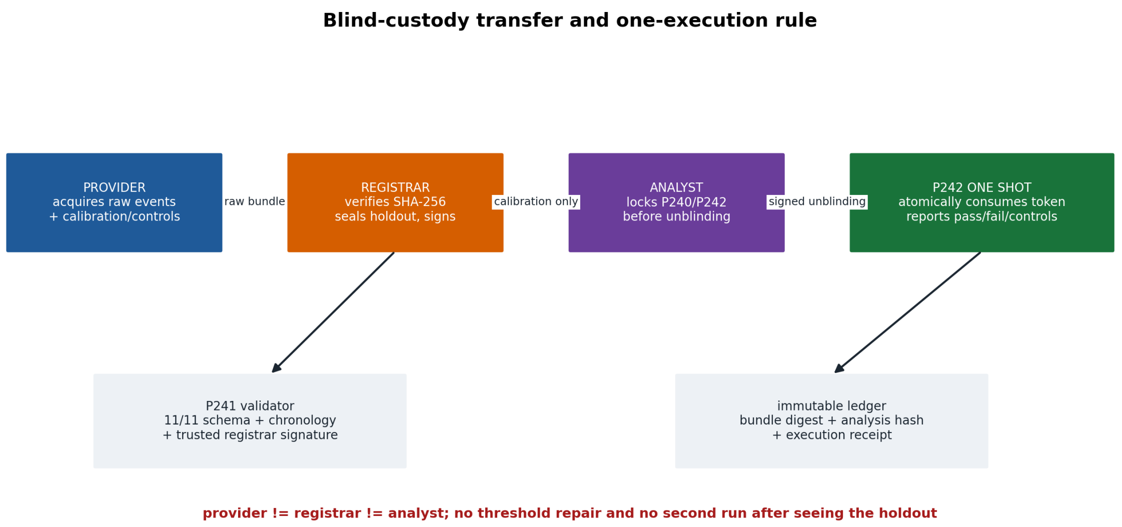

Figure 14.1: Provider, registrar and analyst remain distinct. The sealed holdout is released only after hashes and analysis code are frozen.

Chapter 15

Recommended direct experiment

15.1 Experimental object

The cleanest direct target is a paired twelve-state realization:

$$\mathcal{E}_{\text{lab}} = (\mathcal{P}_{12}, C_\tau, \Phi_\tau, \mathcal{M}_{12}, \mathcal{R}, \mathcal{C}_{\text{custody}}),$$

where $\mathcal{P}_{12}$ are twelve basis preparations, C_τ is the calibrated clock, Φ_τ is the realized process, $\mathcal{M}_{12}$ is the resolving measurement, $\mathcal{R}$ is the event record and $\mathcal{C}_{\text{custody}}$ is the independent chain.

15.2 Registered times

Program 240 fixes

$$\tau\lambda_{\max} = 1, \qquad \tau = 0.426952295949885$$

in dimensionless units. The physical duration is determined by the measured rate scale:

$$\tau_{\text{phys}} = \frac{0.426952295949885}{\kappa}$$

for the heat implementation.

The registered process times are:

Split	**Dimensionless time**	**Purpose**
calibration	τ	estimate $\widehat{P}_\tau$
sealed holdout	2τ	test $\widehat{P}_\tau^2$
optional coherent diagnostic	$\tau/4, \tau/2, \tau$	linear-vs-quadratic escape and interference

15.3 Shot plan

The recommended confirmatory allocation is:

$$50\,000 \quad \text{events per preparation per registered heat time.}$$

For twelve preparations and two times this is

$$2 \times 12 \times 50\,000 = 1\,200\,000$$

registered heat-process events, before controls and calibration.

A pilot may use 10 000 events per preparation. It must not replace the confirmatory run after seeing pilot outcomes unless the adaptation was preregistered.

15.4 Randomization

- randomize preparation order in blocks;
- interleave τ calibration runs and control runs;
- keep 2τ labels sealed from the analyst;
- randomize double-slit shutter configurations if P-C is used;
- record every dropped event and dead-time exclusion;
- never reorder the archived raw stream.

15.5 Calibration

The following calibrations are mandatory:

1. physical clock and trigger latency;
2. twelve input preparation confusion matrix;
3. twelve output detector confusion/crosstalk matrix;
4. efficiency per channel;
5. dark-count and afterpulse rates;
6. dead time;
7. timing jitter;
8. coupling/process stability over the run;
9. spatial or mode relabelling map;
10. environmental monitors and background runs.

Calibration parameters may be propagated as nuisance parameters. They may not be fitted separately to the sealed 2τ target.

Chapter 16

Program 240 — optimal spectral tomography

16.1 Exact time optimum

Normalize the fastest nonzero generator rate to one and write

$$x = \tau\lambda_{\max}.$$

For a transition eigenvalue $\mu = e^{-x}$, inverse-log recovery has local absolute-noise amplification

$$g(x) = \frac{e^x}{x}.$$

Since

$$g'(x) = \frac{e^x(x-1)}{x^2},$$

$x = 1$ is the unique positive minimizer. This proves the registered $\tau\lambda_{\max} = 1$ rule in the declared local absolute-noise model.

16.2 Equal preparation allocation

The target and declared risk are invariant under cyclic permutation of the twelve labels. For any allocation, average its twelve cyclic images. The average allocation is uniform. Convexity of the design risk implies that this symmetrization cannot increase risk. Thus a minimax allocation exists with equal shots for all twelve basis preparations.

This theorem does not say that uniform allocation remains optimal after detector asymmetry. Measured efficiency differences should be incorporated as preregistered costs or nuisance parameters.

16.3 Nonasymptotic matrix concentration

For each preparation j, let X_{jr} be a one-hot outcome from $P_\tau(\cdot|j)$ and let m shots be taken in every column. The raw transition error is a sum of independent rectangular matrices

$$\widehat{P} - P = \sum_{j,r} \frac{(X_{jr} - p_j)e_j^{\mathsf{T}}}{m}.$$

Each summand has operator norm at most

$$R \le \frac{\sqrt{2}}{m}.$$

The rectangular variance proxy satisfies

$$v \le \frac{n}{2m}.$$

Matrix Bernstein therefore gives, with probability at least $1 - \alpha$,

$$\|\widehat{P} - P\|_2 \le \sqrt{2v \log \frac{2n}{\alpha}} + \frac{2R}{3} \log \frac{2n}{\alpha} =: \varepsilon.$$

Symmetrization does not increase the operator error. If $\varepsilon < \mu_{\min}(P)$, Weyl's theorem and the Lipschitz bound for the matrix logarithm give

$$\|\widehat{A} - A\|_2 \lesssim \frac{2\varepsilon}{\tau(\mu_{\min} - \varepsilon)}.$$

If this absolute rate error is $\delta < \lambda_{\max}$, the normalized fingerprint error is bounded by

$$\frac{2\delta}{\lambda_{\max} - \delta}.$$

The bound is deliberately distribution free and conservative.

16.4 Executed planning result

At $\tau\lambda_{\max} = 1$:

$$\lambda_{\min}(P_\tau) = e^{-1} = 0.3678794411714421,$$

and

$$\lambda_{\min}(P_{2\tau}) = e^{-2} = 0.1353352832366125.$$

Three hundred deterministic synthetic trials per shot level gave:

Shots per preparation	**Mean projective distance**	**95th percentile**	**Raw 0.02 pass rate**
10,000	0.016629	0.026915	0.7233
25,000	0.009632	0.015303	0.9900
50,000	0.006419	0.010079	1.0000

The distribution-free matrix-Bernstein/log sufficient count for a guaranteed 0.02 projective bound at $\alpha = 0.05$ is

$$22\,565\,113$$

shots per preparation. This is not the recommended practical count. It is an honest statement of the price of a broad worst-case guarantee. The 50 000 plan is supported by target-model simulation and must be accompanied by the finite-count primary test and external controls.

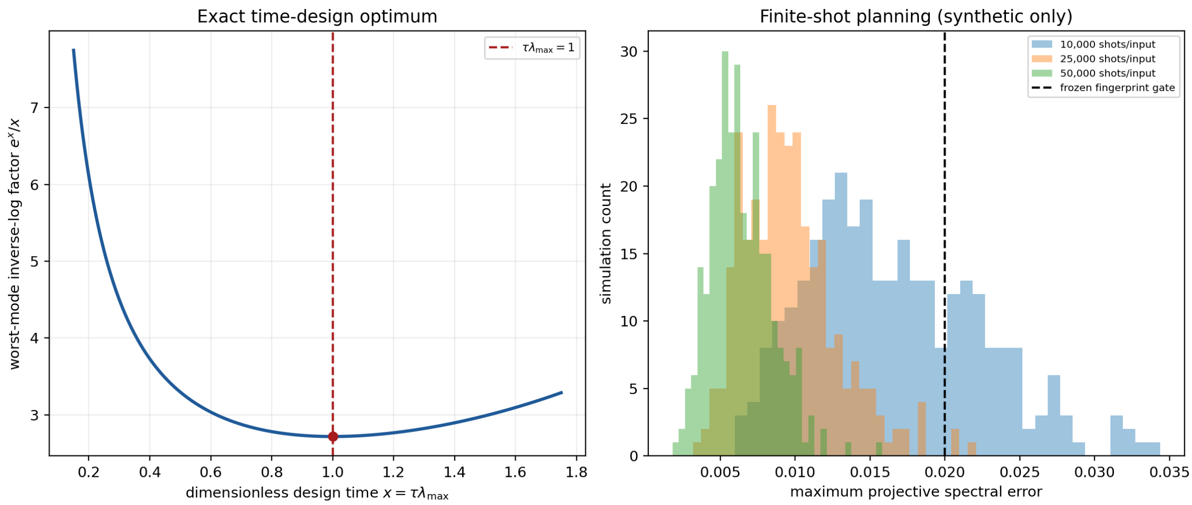

Figure 16.1: Program 240: the exact fastest-mode time optimum and deterministic finite-shot planning distributions.

Chapter 17

Program 241 — executable validator

17.1 Eleven fields

The validator checks:

1. independent provider identity and license;
2. immutable raw-event hashes;
3. ordered events or detection coordinates;
4. preparation or slit-configuration labels;
5. intervention or shutter labels;
6. clock and dimensional calibration;
7. detector geometry, efficiency, dark-count and blur calibration;
8. environment and background record;
9. negative controls;
10. held-out target committed before analysis;
11. provider–registrar–analyst role separation plus trusted registrar signature.

17.2 Bundle layout

BUNDLE/ bundle_manifest.json bundle_manifest.json.asc events.csv calibration.json controls.json environment.json preregistration.json

bundle_manifest.json contains the SHA-256 of every declared file. Paths may not escape the bundle and symlinks are forbidden.

17.3 Process event schema

event_id timestamp_utc run_id subset preparation_id outcome_id evolution_multiple intervention

The production semigroup bundle must contain all preparations and outcomes $0, \ldots, 11$, evolution multiples 1 and 2, and the subsets calibration and holdout.

17.4 Double-slit event schema

event_id timestamp_utc run_id subset configuration x_detector y_detector intervention

The required configurations are both_open, left_only, right_only and both_closed.

17.5 Cryptographic boundary

P241 requires a detached GPG signature of bundle_manifest.json verified against a separately supplied trusted registrar keyring. This establishes file-level provenance relative to that key. It does not establish that the named people are genuinely independent. The collaboration must audit that fact institutionally.

17.6 Commands

Create empty templates:

python3 fin_lab_p241_validator.py
–emit-template FIN_Lab_P241_Transfer_Template_NEW

Validate a production bundle:

python3 fin_lab_p241_validator.py BUNDLE
–signature BUNDLE/bundle_manifest.json.asc
–trusted-keyring registrar-trustedkeys.gpg
–output admission_certificate.json

Without all eleven fields and the trusted signature, the exit status is nonzero and P242 remains locked.

Chapter 18

Program 242 — one-shot analysis pipeline

18.1 Fail-closed prerequisites

P242 requires:

- exact canonical analysis-lock file;
- P241 signed 11/11 admission;
- resource type heat_process;
- complete twelve-state τ calibration record;
- complete twelve-state sealed 2τ holdout;
- preregistration committing the exact SHA-256 of the P242 analysis lock;
- a registrar-controlled one-run ledger.

A standard double-slit bundle fails the semigroup-readiness check by design.

18.2 Finite-count primary threshold

For an n-category empirical distribution with m samples, the Bretagnolle–Huber–Carol bound is used in the form

$$\|\widehat{p} - p\|_1 \leq \sqrt{\frac{2\,(n \ln 2 + \ln(1/\alpha))}{m}}$$

with the failure budget unioned across twelve columns and the two registered times. For column-stochastic matrices,

$$\|\widehat{P}_\tau^2 - P_\tau^2\|_1 \leq 2\|\widehat{P}_\tau - P_\tau\|_1.$$

At 50 000 shots per preparation at both times, the registered maximum column-TV radius is

$$0.0361142590.$$

The labels are:

- FALSIFIED_AT_REGISTERED_LEVEL if the observed statistic exceeds the registered bound;

- NOT_FALSIFIED_AT_REGISTERED_LEVEL otherwise.

The second label is intentionally not VALIDATED.

18.3 Negative controls

P242 reports:

1. cyclic preparation-label shift;
2. reversed/misassigned time;
3. a nearest-neighbour C_{12} heat model whose single scale is fitted only on calibration data and then evaluated on the holdout.

The collaboration should additionally report apparatus-specific controls, missing events and all exclusion counts.

18.4 Atomic execution

Before unblinded analysis, P242 creates

FIN_P242_EXECUTED_<bundle_digest>.lock

with an exclusive filesystem operation. A second attempt with the same bundle digest is refused.

18.5 Production command

python3 fin_lab_p242_pipeline.py
–bundle BUNDLE
–signature BUNDLE/bundle_manifest.json.asc
–trusted-keyring registrar-trustedkeys.gpg
–ledger-dir REGISTRAR_LEDGER
–output FIN_P242_external_result.json

No such command is executed in this release because no external signed bundle is supplied.

Chapter 19

Statistical interpretation

19.1 Registered hypotheses

The primary null is conditional:

$$H_{\mathrm{SG}}: \quad P_{2\tau} = P_\tau^2 \quad \text{for the realized, calibrated, time-homogeneous process.}$$

The FIN fingerprint hypothesis is:

$$H_{\mathrm{FIN}}: \quad \mathrm{spec}_+(A)/\lambda_{\max} \quad \text{matches the frozen strict target.}$$

These are different hypotheses. A generic time-homogeneous Markov process may satisfy H_{SG} and fail H_{FIN}.

19.2 Decision table

Semigroup	**Fingerprint**	**Interpretation**
fail	any	registered time-homogeneous heat realization is falsified or apparatus nonstationarity/calibration failure occurred
not falsified	fail	a semigroup-like process exists but not the frozen strict fingerprint
not falsified	pass	registered implementation survives; compare alternatives and engineering provenance
pass only after refit	invalid	post-hoc repair; report original failure

19.3 Emulator boundary

If the apparatus was programmed from the FIN matrix, a successful run shows that the matrix, control stack and analysis can be implemented. It is not an independent discovery of the matrix in nature. A stronger physical claim requires a domain selected independently of the FIN fit and competitive held-out performance against alternative models.

19.4 Alternative models

At minimum compare:

- nearest-neighbour C_{12} heat flow;
- generic symmetric circulant generator with preregistered parameter count;
- generic reversible Markov model;
- coherent continuous-time quantum walk;
- dephasing/open-system alternatives;
- detector-confusion and clock-drift nuisance models.

Model flexibility must be penalized and all nuisance fitting must use calibration data only.

Chapter 20

Double-slit relation to FIN duality

The finite FIN two-path calculation on $\mathbb{Z}_{12}$ uses a preparation

$$\frac{|a\rangle|E_a\rangle + e^{i\phi}|b\rangle|E_b\rangle}{\sqrt{2}}$$

and environment overlap

$$\eta = \langle E_b|E_a\rangle.$$

After U_t, a vertex detector gives

$$p_{\phi,\eta}(x,t) = \frac{|u_a|^2 + |u_b|^2}{2} + \mathrm{Re}\left(\eta e^{-i\phi} u_a \overline{u_b}\right).$$

This formula demonstrates how preparation, environment record and detector enter an operational prediction. It is not a derivation of the continuum optical diffraction law.

The proposed physical double-slit setup P-C tests the same general operational distinction—coherent cross terms versus controlled one-slit and dark records—but not the FIN spectral fingerprint unless a separate, pre-frozen twelve-mode encoding is supplied.

The double-slit record is therefore valuable for:

- validating event-level acquisition;
- measuring how interference builds from individual detections;
- testing shutter and dark controls;
- exercising provider/registrar/analyst custody;
- demonstrating that a final image is not a sufficient scientific record.

It is not, by itself, the P242 semigroup experiment.

Chapter 21

Apparatus acceptance tests

Before evidence acquisition, the experimental team should demonstrate:

21.1 Source

- stable event rate;
- measured single-photon or weak-source statistics;
- herald timing where applicable;
- no preparation-dependent source drift beyond preregistered tolerance.

21.2 Preparation

- each of twelve inputs addressable;
- confusion matrix measured;
- no omitted input;
- randomized run order.

21.3 Dynamics

- coherent arm: independently reconstructed unitary or Hamiltonian within tolerance;
- heat arm: independently reconstructed rate/process matrix;
- stationarity across run blocks;
- explicit evidence that dephasing/loss is not being confused with the target Markov channel.

21.4 Measurement

- twelve distinct outcomes;
- detector efficiency, dark count, crosstalk and dead-time record;
- position/vertex labels fixed before analysis;
- raw timestamps retained.

21.5 Clock

- time base traceable to an external standard;
- trigger offsets and jitter measured;

- mapping from physical time to dimensionless τ frozen.

21.6 Custody

- three distinct role identities;
- registrar key established before data transfer;
- every raw file hashed;
- holdout sealed;
- analysis-lock hash in preregistration;
- one-execution ledger outside analyst control.

Chapter 22

Falsification and stop rules

The following outcomes must stop claim promotion:

1. any required P241 field fails;
2. registrar signature fails;
3. a file hash changes;
4. preparation or outcome coverage is incomplete;
5. the τ and 2τ clocks are not commensurately calibrated;
6. process drift exceeds the preregistered stability tolerance;
7. the empirical transition spectrum is nonpositive where the logarithm is required;
8. T_{CK} exceeds the registered finite-count radius;
9. the fingerprint fails 0.02;
10. a flexible alternative explains the holdout at least as well after complexity control;
11. the run requires a threshold, label or model change after unblinding.

A failed implementation may motivate a new experiment. It does not permit a second P242 run on the same bundle.

Chapter 23

What a successful experiment would and would not show

23.1 It would show

- a real apparatus can realize the declared twelve-state process;
- the event-level data satisfy the registered semigroup consistency test;
- the projective spectrum agrees or disagrees with the strict finite target;
- coherent and diffusive operational protocols can be compared under explicit preparations, clocks and instruments;
- the transfer/custody procedure is reproducible by independent teams.

23.2 It would not show

- that the strict kernel is uniquely selected by fundamental physics;
- that the legacy kernel is physically transferred to the strict kernel;
- that the dimensional constants emerge from FIN;
- that an absolute orientation or selector emerges;
- that a continuum, Lorentz symmetry, gauge sector, Standard Model or gravity has been derived;
- that FIN is a Theory of Everything.

The strongest permissible positive conclusion after an emulator success is:

> *The frozen finite FIN operator and its registered operational heat/coherent tests were implemented and survived the declared held-out analysis on this apparatus.*

Chapter 24

Handoff responsibilities

24.1 Theoretical physicist

- verify the matrix convention and spectral target;
- inspect the coherent and Lindblad/stochastic realizations;
- define admissible nuisance models;
- check that the physical channel corresponds to the declared mathematical object.

24.2 Experimental physicist

- select hardware and characterize limitations;
- establish physical time and detector calibration;
- validate state preparation and measurement;
- preserve raw events and stability monitors.

24.3 Statistician / analyst

- freeze the analysis hash;
- inspect P240 bounds and practical power;
- preserve the registered failure budget;
- report alternatives and non-rejections correctly.

24.4 Independent registrar

- maintain the trusted signing key and ledger;
- verify chronology and hashes;
- seal/unseal the holdout;
- prevent repeated analysis of the same bundle.

24.5 Repository maintainer

- preserve source, tests, locks, manifest and PDF hashes;
- do not replace external events with generated examples;
- do not change P242 after the preregistration hash is published.

Chapter 25

Executable inventory

File	Function
fin_lab_p240_optimal_tomography.py	target construction, time/allocation theorem, concentration bounds and synthetic planning
FIN_Lab_P240_Design_Lock.json	canonical registered design
FIN_Lab_P240_Optimal_Tomography_Results.json	machine-readable executed P240 results
fin_lab_p241_validator.py	eleven-field schema/hash/chronology/ signature validator
FIN_Lab_P241_Transfer_Template/	empty transfer templates; not evidence
fin_lab_p242_pipeline.py	fail-closed one-run external analysis
FIN_Lab_P242_Analysis_Lock.json	canonical frozen P242 analysis
test_fin_lab_p240_242.py	regression/falsification tests
fin_lab_transfer_figures.py	code-native scientific diagrams

The laboratory should archive the exact SHA-256 manifest delivered with this PDF.

Chapter 26

Reproduction commands

Run P240:

MPLCONFIGDIR=/tmp/matplotlib-finlab
python3 fin_lab_p240_optimal_tomography.py –replicates 300

Regenerate the analysis lock:

python3 fin_lab_p242_pipeline.py –write-lock-only

Run the tests:

MPLCONFIGDIR=/tmp/matplotlib-finlab
python3 -m unittest -v test_fin_lab_p240_242.py

Expected result:

Ran 18 tests OK

The P242 production command must not be run until P241 has admitted a genuine external bundle.

Chapter 27

Literature and methodological anchors

The apparatus proposals use established frameworks. Their use here does not claim novelty for those frameworks.

1. E. Farhi and S. Gutmann, "Quantum computation and decision trees," *Physical Review A* **58**, 915–928 (1998). https://doi.org/10.1103/PhysRevA.58.915
2. H. B. Perets et al., "Realization of quantum walks with negligible decoherence in waveguide lattices," *Physical Review Letters* **100**, 170506 (2008). https://doi.org/10.1103/PhysRevLett.100.170506
3. W. R. Clements et al., "Optimal design for universal multiport interferometers," *Optica* **3**, 1460–1465 (2016). https://doi.org/10.1364/OPTICA.3.001460
4. F. Caruso et al., "Fast escape of a quantum walker from an integrated photonic maze," *Nature Communications* **7**, 11682 (2016). https://doi.org/10.1038/ncomms11682
5. P. Kolenderski et al., "Time-resolved double-slit interference pattern measurement with entangled photons," *Scientific Reports* **4**, 4685 (2014). https://doi.org/10.1038/srep04685
6. F. A. Pollock et al., "Non-Markovian quantum processes: Complete framework and efficient characterization," *Physical Review A* **97**, 012127 (2018). https://doi.org/10.1103/PhysRevA.97.012127
7. P. Taranto et al., "The structure of quantum stochastic processes with finite Markov order," *Physical Review A* **99**, 042108 (2019). https://doi.org/10.1103/PhysRevA.99.042108
8. J. A. Tropp, "User-friendly tail bounds for sums of random matrices," *Foundations of Computational Mathematics* **12**, 389–434 (2012). https://doi.org/10.1007/s10208-011-9099-z
9. T. Weissman et al., "Inequalities for the L1 deviation of the empirical distribution," Hewlett-Packard Laboratories Technical Report HPL-2003-97 (2003).
10. B. Bylander et al., "Noise spectroscopy through dynamical decoupling with a superconducting flux qubit," *Nature Physics* **7**, 565–570 (2011). https://doi.org/10.1038/nphys1994

Chapter 28

Final transfer statement

The finite mathematical result is strong and narrow:

$$A = sI - W$$

is a positive connected weighted-graph Laplacian for the frozen strict twelve-node FIN profile, and the same spectrum supports coherent and diffusive functional calculi.

The missing bridge is not another formula in A. It is an independently implemented and calibrated operational process with a real record.

This release supplies the part that can honestly be supplied before a laboratory acts:

$$\boxed{\text{P240 design lock} \ + \ \text{P241 signed validator} \ + \ \text{P242 one-shot pipeline}}$$

The fifth duty remains deliberately empty. Only an independent experimental collaboration can fill it.

Release reproducibility

```
MPLCONFIGDIR=/tmp/matplotlib-finlab \
python3 fin_lab_p240_optimal_tomography.py --replicates 300

python3 fin_lab_p242_pipeline.py --write-lock-only

MPLCONFIGDIR=/tmp/matplotlib-finlab \
python3 -m unittest -v test_fin_lab_p240_242.py
```

Expected test result: 18 tests, all passing. No P242 production execution is part of this release because no signed external P241 bundle is supplied.

Suggested citation:

Żuchowski, K. (2026). *FIN Laboratory Transfer Package: P240 Optimal Tomography, P241 Blind-Custody Validator, and P242 One-Shot Pipeline* (Release 10.21; Version 1.0.0) [Preprint and executable specification]. Independent Researcher — Fractal Information Theory Project.